

Name of Program	Bachelor of Technology (Computer Science Engineering)				
CSET109	Object Oriented Programming using Java	L	T	P	C
Owning School/Department	School of Computer Science Engineering and Technology	3	1	4	6
Pre-requisites/Exposure	-				

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To examine different programming structures in a platform independent language such as wrapper classes, collections, exceptions, and multithreading.

CO2: To explain the concepts of object-oriented programming like encapsulation, abstraction, inheritance and polymorphism.

CO3: Make use of GUI and database-based programming to develop Applications for real life problems.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3		3	3								3		
CO2	3	3		3	3								3	3	3
CO3			3		3								3	3	3

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

10 lecture hours

Handling user input/output using Scanner class, Platform independence of Java (JVM), Operators (unary, arithmetic, logical, shift left, shift right, ternary, assignment) Exploring bitwise operators, compound assignment, if/else, Using switch for menu-driven applications while, For, do-while, Introduction to Abstraction and Polymorphism Objects Lifecycle: creation, dereference, garbage collection, Wrapper classes (Boolean, Integer, Double, Character), Autoboxing, unboxing, String (concat, indexOf, split, length, toLowerCase, toUpperCase, replace, trim), Array declaration, instantiation, multidimensional arrays, Designing methods for modularity and reusability Static variables, Static methods, Static blocks, Constructors and destructors, Instance Initialization Blocks, Security aspects in class design Encapsulation, Inheritance, single, multilevel, Hierarchical.

Module II:

11 lecture hours

Polymorphism, Function Overloading method overriding, Concrete Class Abstraction abstract methods, non-availability of multiple inheritance, interfaces, Interfaces vs abstract class, Embedded Interface Anonymous class, Inbuilt Packages, User defined packages, Array List collection, LinkedList Collection Vector collection, Exception handling, Checked and unchecked exception, try, catch, finally, Propel, Propagate, Data Stream Handling.

Module III:

11 lectures hours

Thread and process, Parallel Computing Concurrent Programming synchronization, Swing features, JavaFX Features MySQL database, NoSQL Databases getConnection, createStatement, executeQuery, JEE (client-server architecture for web based applications), Microservices Architecture Running Servlet, Generic Servlet, HTTP Servlet, Servlet Config, Web Filters Servlet to handle Get and Post Methods, Startups on programming, Session Management.

Module IV:**10 lectures hours**

JSPs, Struts framework, Spring MVC framework Regular expressions (Lambda expressions), Collection framework Concurrency in Java, HashMap, Linked Hash Map, Tree Map, Kotlin for Android WebSockets JPA (Java Persistence API), Microservices with Spring Boot.

Studio Work / Laboratory Experiments:

Students will gain hands-on experience on core JAVA. Students will be able to solve simple to medium level computation problems in Java. Emphasis will be that the student code themselves as much as they can. They learn to debug the programs resulting in error free code. Instructor will be giving assignments based on the content covered in the lecture classes in the corresponding week.

Text Books :

1. Schildt, Herbert. *Java: the complete reference*. 10th ed. McGraw-Hill Education Group, 2014. ISBN 978-93392120.
2. Bloch, Joshua. *Effective java (the java series)*. 1st ed. O'Reilly Media, Inc., 2017. ISBN 978-0134686097.

Reference Books :

1. Anuradha A. Puntambekar, *ObjectOriented Programming*. 1st ed. UNICORN Publishing Group, 2020. ISBN 9789333223819.